107	+12V	-	108	+12V	-
109	+12V	-	110	+12V	-
111	+12V	-	112	+12V	-
113	+12V	-	114	+12V	-

115	+12V	-	116	+12V	-
117	+12V	-	118	+12V	-
119	+12V	-	120	+12V	-

2.

Expansion board

2.1

Introduction

Through the previous function introduction, we can understand the functions of the expansion board

- 1 channel M.2 interface
- 1 channel DP output interface
- 4 USB 3.0 ports
- 2 Gigabit Ethernet interfaces
- 2 USB Uart ports
- 1 way Micro SD card holder

- 1 channel MIPI camera interface
- 1 channel HDMI video output interface
- 1 channel HDMI video input interface
- 1 way FMC expansion port
- 2 CAN communication interfaces
- 2-Channel 485 communication interface
- JTAG debugging port
- 1-Channel temperature sensor
- 1 way EEPROM
- 1 RTC real-time clock;

3 LED lights

-

3 buttons

lways @(posedge clk) egin <= d ; nd ndmodule	

Combinational logic can also be generated, such as data selectors. Sensitive signals do not have clocks, and reg Mux is defined. The final generated circuit is combinational logic.

```
odule top(a, b, c, d, sel, Mux) ;

put a ;

put b ;

put c ;

put d ;

put [1:0] sel ;

utput reg Mux ;



lways @(sel or a or b or c or d)

egin
```

```
ase(sel)

'b00 : Mux = a ;

'b01 : Mux = b ;

'b10 : Mux = c ;

'b11 : Mux = d ;

ndcase

nd

ndmodule
```

3.3

Memory type

Memory types such as RAM and ROM can be defined with the structure of reg [n-1:0] memory name [m-1:0], which means m registers with n-bit width. For example, reg [7:0] ram [255:0] represents 256 8-bit registers defined, where 256 is the depth of the memory and 8 is the data width.